Insulation Trial: Trace the Heat

Tracking #: 2026-11-14-NAT-SCI-INSULATION-47

Fictional 15-minute cup cooling data; no hot water is handled

Cup	Start temp	End trials 1 / 2 / 3
A: insulated	40 C	33, 32, 34 C
B: unwrapped	40 C	30, 29, 31 C
Kept same	Volume, room, wait	15 minutes each

1. Draw the energy-transfer story from warmer cup toward cooler room. Label the stored thermal energy, surroundings and the role of insulation in TWO sentences.

2. Write an IV/DV/control table. Name the single changed feature, measured outcome and THREE factors kept identical.

Insulation Trial: Trace the Heat

Tracking #: 2026-11-14-NAT-SCI-INSULATION-47

Fictional 15-minute cup cooling data; no hot water is handled

Cup	Start temp	End trials 1 / 2 / 3
A: insulated	40 C	33, 32, 34 C
B: unwrapped	40 C	30, 29, 31 C
Kept same	Volume, room, wait	15 minutes each

3. Calculate all three temperature drops for cup A and then its mean drop. Show 40 minus end temperature and divide by three.

4. Do the same for cup B. Compare the two mean drops and quantify the difference in degrees Celsius.

Insulation Trial: Trace the Heat

Tracking #: 2026-11-14-NAT-SCI-INSULATION-47

Fictional 15-minute cup cooling data; no hot water is handled

Cup	Start temp	End trials 1 / 2 / 3
A: insulated	40 C	33, 32, 34 C
B: unwrapped	40 C	30, 29, 31 C
Kept same	Volume, room, wait	15 minutes each

5. A student says ’the data prove cup A stored exactly 7 joules more energy.’ Explain in THREE sentences why the units and evidence do not support that claim.

6. Write a six-sentence lab memo with experimental question, controls, both means, difference, a careful conclusion and one limitation of fictional data.